



EDGEN MURRAY

an EDGEN Company

Bill of Lading

Reference No. : 385059-1

Shipment Date : 4/7/2025

From : Edgen Murray Corporation, P.O. Box 84160, Baton Rouge, LA 70884-4160

Ship To : CONSUMERS ENERGY COMPANY, MILLER TRUCKING COMPANY, 1800 MITCHELL, Jackson, MI 49203
(517) 788-8392

Terms Of Trade : FOB: Shipping Point

Delivery Terms : 3rd Prty Collect

Tracking No : 4500953195

Carrier : ACF GLOBAL LOGISTICS

Vehicle No :

Trailer No :

11 PCS (Valve)

Total : 11 PCS 1,034 LBS

Attachments : PKL-385059-1

NMFC #52190-6 CLASS: 50 (UNLESS OTHERWISE NOTED)

SIGNATURE FOR GOODS RECEIVED: _____

PRINT NAME: _____ DATE: _____

The property described above, in apparent good order, except as noted (contents and condition of contents of packages unknown), marked, consigned, and destined as indicated below, which said carrier (the word carrier being understood throughout this contract as meaning any person or corporation in possession under the contract) agrees to carry to its usual place of deliver at said destination, if on its route, otherwise to deliver to another carrier on the route to said destination, it is mutually agreed, as to each carrier of all or any said property over all or any portion of said route to destination, and as each party at any time interested in all or any of said property, that every service to be performed hereunder shall be subject to all the terms and conditions of the Uniform Domestic Straight Bill of Lading set forth (1) in Official, Southern, Western and Illinois Freight Classifications in effect on the date hereof, if this is a rail or a rail-water shipment, or (2) in the applicable motor carrier classification or tariff if this is a carrier shipment.

Shipper hereby certifies that he is familiar with all the terms and conditions of the said bill of lading, including those on the back thereof, set forth in the classification or tariff which governs the transportation of this shipment, and the said terms and conditions are hereby agreed to by the shipper and accepted for himself and his assigns.

Subject to Section 7 of the Conditions of applicable bill of lading, if this shipment is to be delivered to the consignee without recourse on the consignor, the consignor shall sign the following statement: The carrier shall not make delivery of this shipment without payment of freight and all other lawful charges		CARRIER	FREIGHT Prepaid [] Collect []
SIGNATURE OF CONSIGNOR X		AGENT PER(Driver's Signature) X	
If charges are to be prepaid, write or stamp here, "To be Prepaid."	RECEIVED To apply in payment of the charges on the property described hereon. \$	AGENT OR CASHIER (The Signature here acknowledges only the amount prepaid.)	CHARGES ADVANCED \$

PREPAID FREIGHT:

All LTL Invoices bill to: Edgen Murray Corp, C/O Cerasis, Inc., P.O. Box 21248, Eagan, MN 55121-0248.

All OTHER carriers (TL/Dedicated/Parcel), SEND TO: Edgen Murray Corp, P.O. Box 84160, Baton Rouge, LA 70884-4160

4/7/2025

04:19 PM



1



EDGEN MURRAY

an EDGEN Company

Packing List

Packing List : 385059-1
Ship Date : 4/7/2025
Cust/Ship to: L12187/2

Ship-To :
CONSUMERS ENERGY COMPANY
MILLER TRUCKING COMPANY
1800 MITCHELL
Jackson, MI 49203

Issued from :

DIRECT SHIPMENTS
DIRECT SHIP SEE SHIP ADDRESS IN REMARKS BELOW

Delivery Terms : 3rd Prty Collct
Terms of Trade : FOB: Shipping Point
Carrier : ACF GLOBAL LOGISTICS

Vehicle No :
Trailer No :
Tracking No : 4500953195

Our Order : 2485574 **Your PO :** 4500953195 (2/20/2025)

Item Product

1-1

**2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE
PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT
WITH DB-2 WRENCH OPERATOR**

Mill/Heat
FLV/COC

Tag/ID
314001-1

Qty

Pcs
11

Total
1,034.00

Total for the Shipment **11 PCS 1,034 LBS**

Salesperson : Nick Polizzi (314) 524 0001
Expediter : Margaret Taylor (314) 524 0001
Cus Contact :

nick.polizzi@edgenmurray.com
maggie.taylor@edgenmurray.com

We appreciate any feedback from you concerning quality and service of all delivered goods. This will enable us to establish a customer satisfaction survey and to take action for continuous quality improvement.

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR



Experience In Motion

PACKING SLIP

SHIP TO:
 CONSUMERS ENERGY
 COMPANY
 1800 MITCHELL ST
 MILLER TRUCKING
 JACKSON MI 49203
 AMERICA

Customer number:	36102
Date:	07-APR-2025
Contact:	Lindburg, Rhonda
Bill of Lading No.:	92229
Packing List No.:	167985
Freight Terms:	Third Party Billing
Shipping Methods:	ABF-LTL-Standard
Payment Terms:	Net 45
Shipping Instructions:	

INVOICE TO:
 EDGEN MURRAY CORPORATION
 BATON ROUGE LA 70884-4160
 AMERICA

FLOWSERVE ORDER NO:

CUSTOMER REFERENCE NO:

1113048

STL-314001



Line No	Sales Order Line No	Part No.	Customer Part No. / Description	Qty	UOM	Custom tariff Number	Net Weight (LBS)
1	1.1	F01652349-1033	2 IN K2345 VALVE DM W/1033 - W/1033 SEALANT Country Of Origin: India	3	Each	8481803065 EAR99	0
2	1.1	F01652349-1033	2 IN K2345 VALVE DM W/1033 - W/1033 SEALANT Country Of Origin: India	5	Each	8481803065 EAR99	0
3	1.1	F01652349-1033	2 IN K2345 VALVE DM W/1033 - W/1033 SEALANT Country Of Origin: India	3	Each	8481803065 EAR99	0
4	2.1	00482006	DB2 WRENCH ASSEMBLY - DB2 WRENCH ASSEMBLY Country Of Origin: India	11	Each	8204110030 EAR99	0

Total parcels: 1
Total weight: 1256 LBS
Dimensions:

Flowserve US Company
 1511 JEFFERSON STREET
 SULPHUR SPRINGS, TX 75482
 UNITED STATES
 MAIN TEL.: +1 903 885 3151
 NORDSTROM FAX: +1 903 439 3444
www.flowserve.com

Valbart Sales:
 Phone : +1 281 671 1025
 Fax: +1 281 671 1609



Experience In Motion

PACKING SLIP

SHIP TO:
CONSUMERS ENERGY
COMPANY
1800 MITCHELL ST
MILLER TRUCKING
JACKSON MI 49203
AMERICA

Customer number:	36102
Date:	07-APR-2025
Contact:	Lindburg, Rhonda
Bill of Lading No.:	92229
Packing List No.:	167985
Freight Terms:	Third Party Billing
Shipping Methods:	ABF-LTL-Standard
Payment Terms:	Net 45
Shipping Instructions:	

INVOICE TO:
EDGEN MURRAY CORPORATION
BATON ROUGE LA 70884-4160
AMERICA

NOTICE: It is your obligation to comply with all applicable US and/or local country export, re-export, and/or economic sanctions laws and regulations and to obtain ANY necessary government approval prior to the export and/or reexport of Flowserve products and/or technology. Please be aware that diversion of Flowserve products contrary to US and/or local country law is strictly prohibited and that virtually all exports AND reexports OF US-ORIGIN PRODUCTS AND/OR TECHNOLOGY AS WELL AS other transactions with certain countries, including without limitation, Cuba, North Korea, Iran, Sudan and Syria are strictly prohibited. Failure to comply with all applicable US and/or local country export, reexport and/or economic sanctions laws and regulations may result in serious consequences including, without limitation, monetary fines, loss of export privileges, imprisonment and/or blacklisting by U.S. and/or local governments. For more information contact your local Flowserve Customer Service Representative.

Certificate of Conformance:

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve US Company specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of Flowserve US Company drawings, ASTM specifications, API Q1 9th Edition, and API 6D 25th edition (if applicable to the API monogram), and ISO 9001:2015 (if applicable). All valves/parts/sealant meet all the requirements of the customer purchase order and EN10204 (when applicable). Flowserve Dynamic Balanced Valves are provided with a pressure balance feature unless specifically identified in the Valve description.

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR



Monday Apr-07-2025

Flowserve Facility:

Flowserve Nordstrom / Valbart Valves
1511 Jefferson St.
Sulphur Springs, TX 75482

cmtrs@flowserve.com

To:***CERTIFICATE OF CONFORMANCE***

SERIAL #	HEAT #	PO #	Description	Class	End Connection
61-693996	T0234	STL-314001	2" 2345	900	FLG
61-693997	T0234	STL-314001	2" 2345	900	FLG
61-693998	4859	STL-314001	2" 2345	900	FLG
61-685273	J7842	STL-314001	2" 2345	900	FLG
61-685268	K7939	STL-314001	2" 2345	900	FLG
61-685266	J7872	STL-314001	2" 2345	900	FLG
61-685267	J7809	STL-314001	2" 2345	900	FLG
61-685262	J7586	STL-314001	2" 2345	900	FLG
61-685263	J7571	STL-314001	2" 2345	900	FLG
61-685261	J7494	STL-314001	2" 2345	900	FLG
61-685265	S4501	STL-314001	2" 2345	900	FLG

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (If Applicable), API Q1 10th edition, and API 6D 25th edition.(if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically identified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

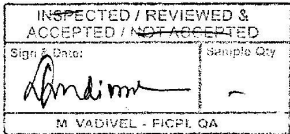
A handwritten signature in black ink, appearing to read "J. McKee", is written over a horizontal line.

FORM QF-MT VER 4

Documentation Technician
Justin McKee

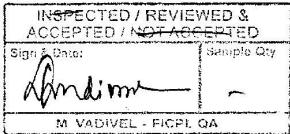
Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

 SAMCO METALS AND ALLOYS PVT.LTD. ISO 9001 - 2015. TUV CERT.REG.NO:44 100 094292 AD-WO TUVNORD APPROVAL KANIYAMBADI,VELLORE Pin :632 102, Phone :0416-2230221, 2230222, 2917614, Fax :0416-2230459, email :samco@samcoalloys.in INDIA. CERTIFICATE OF MATERIAL AND TESTING FOR STEEL CASTING (As per EN 10204-3.1)															
Customer : FLOWSERVE INDIA CONTROLS PVT.LTD B8, MMDA INDUSTRIAL AREA , MARAIMALAI NAGAR -603209										T.C. No. : T0234/37 T.C.Date : 24/09/2024					
Reference. 10119784 Dt. 18/08/2024										Melt No. : T0234 Date of Pouring : 31/08/2024					
Specification : ASTM A216/A216M-2021 GR : WCC & RMC 90012 REV.15															
MELTING PROCESSED : BY BASIC INDUCTION SM :MANUFACTURER'S Symbol															
CHEMICAL COMPOSITION - %															
Element Requirement	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	N	TRE	CE	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.080	0.015	1.000	0.430	
Actual	0.201	0.974	0.471	0.020	0.008	0.125	0.065	0.035	0.012	0.001	0.036	0.009	0.238	0.401	
CE : C + (Mn / 6) + ((Cr + Mo + V) / 5) + ((Ni + Cu) / 15) RE : (Cr + Ni + Mo + Cu + V)															
HEAT TREATMENT DETAILS										Cycle No		Furnace Code		HT Quenching	
Normalizing : Raised to 940°C soaked for 4 Hours and then air cooled.										B364/24		HTF-B & F46			
MECHANICAL PROPERTIES															
Req- uiments	Yield Strength				Tensile Strength		%Elongation		%Red. of Area	Hard- ness	Bend Test D = 3T at 90 °	Impact Test in (Joules) (10mm X 10mm X 55mm)			
	ksi	Kgf/Sq.mm	ksi	Kgf/Sq.mm	ksi	Kgf/Sq.mm	4D	5D							
	0.2% Proof Stress		1% Proof Stress				-	-	-	HBW		CVN 2mm 45 °			
	Min	40	28.05	---	---	70	49.47	22.00	---	35.00	---				
Max	---	---	---	---	95	66.81	---	---	---	237		1	2	3	Avg
Actual	45	31.41	---	---	77	54.46	34.04	---	56.47	149	-	--	--	--	--
DETAILS OF CASTING POURED															
SL.No	PO SNo.	Description					Drawing No.			Part No.	Quantity	Produced			
1	3	BODY CAST 2" CL 1500 F.NO.3045					487888 REV.39			N487888WCC	10	10			
Remarks : MP No : MPZ8668, MPZ8669, MPZ8670, MPZ8671, MPZ8672, MPZ8673, MPZ8674, MPZ8675, MPZ8676, MPZ8677.The above castings are meet the specification FTDC/001 REV. 7. Furnace Calibration Date : 25.06.2024 & Due Date : 24.06.2025. We certify that the material was manufactured, sampled, tested and inspected in accordance with the material specification and the purchase order and was found to meet the requirements. Castings have been MPI performed as per the FNDE - 002 REV. 09. Results found satisfactory. Visual inspection carried out as per MSS SP 55. No surface irregularities to type - I , Type II-XIIa - 2011. The above Castings are satisfactory Radiation results within limits of acceptable 0.20 µsv/hr.															
We Certify the particular entered herein are correct and the material satisfies the specification . ASTM A216/A216M-2021 GR : WCC & RMC 90012 REV.15															
 CHITRA.V (Manager QA)						 M VADIVEL - FICPL QA					for SAMCO METALS AND ALLOYS PVT.LTD.  JAGANNATHAN.S (Authorized Works Inspector)				

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

 SAMCO METALS AND ALLOYS PVT.LTD. ISO 9001 - 2015. TUV CERT.REG.NO:44 100 094292 AD-WO TUVNORD APPROVAL KANIYAMBADI,VELLORE Pin :632 102, Phone :0416-2230221, 2230222, 2917614, Fax :0416-2230459, email :samco@samcoalloys.in INDIA. CERTIFICATE OF MATERIAL AND TESTING FOR STEEL CASTING (As per EN 10204-3.1)															
Customer : FLOWSERVE INDIA CONTROLS PVT.LTD B8, MMDA INDUSTRIAL AREA , MARAIMALAI NAGAR -603209										T.C. No. : T0234/37 T.C.Date : 24/09/2024					
Reference. 10119784 Dt. 18/08/2024										Melt No. : T0234 Date of Pouring : 31/08/2024					
Specification : ASTM A216/A216M-2021 GR : WCC & RMC 90012 REV.15															
MELTING PROCESSED : BY BASIC INDUCTION SM :MANUFACTURER'S Symbol															
CHEMICAL COMPOSITION - %															
Element Requirement	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	N	TRE	CE	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.080	0.015	1.000	0.430	
Actual	0.201	0.974	0.471	0.020	0.008	0.125	0.065	0.035	0.012	0.001	0.036	0.009	0.238	0.401	
CE : C + (Mn / 6) + ((Cr + Mo + V) / 5) + ((Ni + Cu) / 15) RE : (Cr + Ni + Mo + Cu + V)															
HEAT TREATMENT DETAILS										Cycle No		Furnace Code		HT Quenching	
Normalizing : Raised to 940°C soaked for 4 Hours and then air cooled.										B364/24		HTF-B & F46			
MECHANICAL PROPERTIES															
Req- uiments	Yield Strength				Tensile Strength		%Elongation		%Red. of Area	Hard- ness	Bend Test D = 3T at 90 °	Impact Test in (Joules) (10mm X 10mm X 55mm)			
	ksi	Kgf/Sq.mm	ksi	Kgf/Sq.mm	ksi	Kgf/Sq.mm	4D	5D							
	0.2% Proof Stress		1% Proof Stress		-		-	-	-	HBW	CVN 2mm 45 °				
	Min	40	28.05	---	---	70	49.47	22.00	---	35.00	---				
Max	---	---	---	---	95	66.81	---	---	---	237		1	2	3	Avg
Actual	45	31.41	---	---	77	54.46	34.04	---	56.47	149	-	--	--	--	--
DETAILS OF CASTING POURED															
SL.No	PO SNo.	Description				Drawing No.				Part No.		Quantity	Produced		
1	3	BODY CAST 2" CL 1500 F.NO.3045				487888 REV.39				N487888WCC		10	10		
Remarks : MP No : MPZ8668, MPZ8669, MPZ8670, MPZ8671, MPZ8672, MPZ8673, MPZ8674, MPZ8675, MPZ8676, MPZ8677.The above castings are meet the specification FTDC/001 REV. 7. Furnace Calibration Date : 25.06.2024 & Due Date : 24.06.2025. We certify that the material was manufactured, sampled, tested and inspected in accordance with the material specification and the purchase order and was found to meet the requirements. Castings have been MPI performed as per the FNDE - 002 REV. 09. Results found satisfactory. Visual inspection carried out as per MSS SP 55. No surface irregularities to type - I , Type II-XIIa - 2011. The above Castings are satisfactory Radiation results within limits of acceptable 0.20 µsv/hr.															
We Certify the particular entered herein are correct and the material satisfies the specification . ASTM A216/A216M-2021 GR : WCC & RMC 90012 REV.15															
 CHITRA.V (Manager QA)						 M VADIVEL - FICPL QA				for SAMCO METALS AND ALLOYS PVT.LTD.  JAGANNATHAN.S (Authorized Works Inspector)					



CASTLEGATE

CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED

S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120
Phone :7397161500 04259264166, Mail :info@castlegatellc.com

INSPECTION CERTIFICATE
According to EN10204 - 3.1

ISO 9001 : 2015
CERTIFIED
COMPANY

Customer : FLOWSERVE INDIA CONTROLS PVT LTD
CHENNAI

T.C. No : FLS/4859 - 001
T.C.Date : 14/09/2024

PO.No : 10118677 Dt. 17/06/2024

Heat No : 4859
Pour Date : 28/08/2024

Specification : ASTM A216 GR.WCC[UNS J02503]-2021

CASTING PROCESSED BY INDUCTION MELTING
FURNACE

FOUNDURY IDENTIFICATION : CG

CHEMICAL COMPOSITION %

Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RE	AL	N
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.080	0.015
Actual	0.208	0.999	0.508	0.020	0.009	0.099	0.092	0.052	0.012	0.005	0.413	0.260	0.046	0.011

HEAT TREATMENT :

NORMALISING: Temperature reised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:D0141/24-25Dt.02/09/2024)

MECHANICAL PROPERTIES

Requirements	Yield Strength		Tensile Strength		%Elongation	% Reduction	Hardness	Bend Test	Impact Test			
	kgf/mm ²	Mpa	kgf/mm ²	Mpa	GL :50 mm	In Area	HBW					
	0.2 % Rp		-		-	-	-	-				
Minimum	28.03	275	49.44	485	22	35	---	---				
Maximum	---	---	66.77	655	---	---	237	---	1	2	3	Avg
Actual	32.34	317.26	54.12	530.9	33.84	65.69	167	---	---	---	---	---

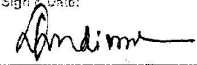
ITEM DESCRIPTION

SL.No.	Description	Drawing No.	Part No	Quantity
1	BODY CSTG 2" CL 1500	487888 REV 39	N487888WCC	9

Heat.SI.No :1,2,3,4,5,6,7,8,9
MP.No : MP24472,MP24473,MP24474,MP24475,MP24476,MP24477,MP24478,MP24479,MP24480 ,

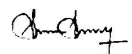
Remarks:
1.This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.
2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.
3.Certified that the Castings are free from Radioactive contaminations.
4.MPI carried out as per MSS SP 53 and found accepted.

INSPECTED / REVIEWED &
ACCEPTED / NOT ACCEPTED

Sign & Date:  Sample Qty: 

M VADIVEL - FICPI QA

We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements


Prepared By
SABARI SANKAR

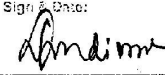

Approved By
ARUN KUMAR.N

CIH/F/QA-04/R0

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

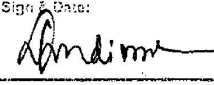
Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:10440935 VSC/LAB/06 - R0			
INSPECTION CERTIFICATE AS PER EN 10204 3.1															
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/J7842/23 T.C.Date : 11-AUG-23					
P.Order : 10112521 Sino 7 Dt. 03/07/2023, 10112911 Sino 3 Dt. 26/07/2023, 10112839 Sino 1 Dt. 19/07/2023, 10112947 Sino 1 Dt. 26/07/2023										Heat No : J7842 Pour Date : 01-AUG-23					
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC					
CHEMICAL COMPOSITION % ()															
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080	
Actual	0.191	0.950	0.421	0.014	0.007	0.099	0.060	0.040	0.007	0.021	0.386	0.227	0.011	0.045	
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 4.00 Hrs and Air Cooled. Cycle No.D9238 Furnace ID.HT/02															
MECHANICAL PROPERTIES ()															
Requirments	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()						
	MPa	Ksi	MPa	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)						
	0.2 % Rp		-		-	-	-								
Minimum	290.00	42	485.00	70	22.00	35.00	---	-							
Maximum	---	---	655.00	95	---	---	237	-	1	2	3	Avg			
Actual	301.49	44	500.14	73	33.92	68.54	143,143,143		--	--	--	--			
ITEM DESCRIPTION															
SL.No.	Description					Drawing No.		Raw Part No/ Finishing Part No		CSL No		Quantity			
1	BODY SN CAST 4" cl 150					2762516 REV 03		N2762516		4, 5		2 Nos			
2	BODY SN CAST 3" CL300					2762446 REV 00		N2762446		6		1 Nos			
3	BODY CAST 3" CI 1500					487879 REV 23		N487879WCC		1		1 Nos			
4	BODY CSTG 2" cl.1500					487888 REV 39		N487888WCC		2, 3		2 Nos			
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to as per ASME B16.34 APPENDIX II. & RMP61158 Rev 4 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.															
INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty:  M VADIVEL - FICPL QA															
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements, material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.															
 M.Kahagalatha, Lab Chemist. Prepared by.										 C. Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT					

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

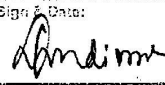
Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:10440935 VSC/LAB/06 - R0		
INSPECTION CERTIFICATE AS PER EN 10204 3.1														
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/K7939/23 T.C.Date : 23-AUG-23				
P.Order : 10112691 Sino 1 Dt. 12/07/2023, 10112727 Sino 1 Dt. 13/07/2023, 10112947 Sino 1 Dt. 26/07/2023										Heat No : K7939 Pour Date : 07-AUG-23				
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC				
CHEMICAL COMPOSITION % ()														
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080
Actual	0.195	0.982	0.447	0.015	0.007	0.099	0.067	0.067	0.010	0.012	0.399	0.255	0.011	0.048
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 7.00 Hrs and Air Cooled. Cycle No.D9254 Furnace ID.HT/02														
MECHANICAL PROPERTIES ()														
Requirments	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()					
	MPa	Ksi	MPa	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)					
	0.2 % Rp		-		-	-	-							
Minimum	290.00	42	485.00	70	22.00	35.00	---	-						
Maximum	---	---	655.00	95	---	---	237	-	1	2	3	Avg		
Actual	306.23	44	505.01	73	34.44	62.48	145,145,143		--	--	--	--		
ITEM DESCRIPTION														
SL.No.	Description				Drawing No.		Raw Part No/ Finishing Part No		CSL No	Quantity				
1	BODY CAST 8" cl 600				122649 REV 28		N122649WCC		1	1 Nos				
2	BODY CAST 6" cl 600				122586 REV 28		N122586WCC		2	1 Nos				
3	BODY CSTG 2" cl.1500				487888 REV 39		N487888WCC		3	1 Nos				
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to as per ASME B16.34 APPENDIX II. & RMP61158 Rev 4 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.														
INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VADIVEL - FICPL QA														
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements, material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.														
 M.Kahagalatha, Lab Chemist. Prepared by.								 C. Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT						

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

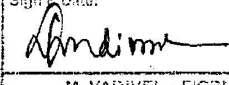
Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:104440935 VSC/LAB/06 - R0			
INSPECTION CERTIFICATE AS PER EN 10204 3.1															
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/J7872/23 T.C.Date : 19-AUG-23					
P.Order : 10112911 Sino 3 Dt. 26/07/2023, 10112521 Sino 5 Dt. 03/07/2023, 10112947 Sino 1 Dt. 26/07/2023										Heat No : J7872 Pour Date : 03-AUG-23					
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC					
CHEMICAL COMPOSITION % ()															
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080	
Actual	0.195	1.013	0.379	0.015	0.008	0.089	0.064	0.036	0.012	0.023	0.399	0.224	0.013	0.052	
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 4.00 Hrs and Air Cooled.Cycle No.B9250 Furnace ID.HT/01															
MECHANICAL PROPERTIES ()															
Requirements	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()						
	MPa	Ksi	MPa	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)						
	0.2 % Rp		-		-	-	-								
Minimum	290.00	42	485.00	70	22.00	35.00	---	-							
Maximum	---	---	655.00	95	---	---	237	-	1	2	3	Avg			
Actual	333.30	48	520.45	75	32.88	59.19	146,146,146		--	--	--	--			
ITEM DESCRIPTION															
SL.No.	Description					Drawing No.		Raw Part No/ Finishing Part No		CSL No		Quantity			
1	BODY CSTG 2" cl.1500					487888 REV 39		N487888WCC		1, 2, 3, 4		4 Nos			
2	BODY SN CAST 3" CL300					2762446 REV 00		N2762446		5		1 Nos			
3	BODY SN CAST 3" CL300					2762446 REV 00		N2762446		6		1 Nos			
4	BODY SN CAST 2" cl 300					2765234 REV 03		N2765234		7		1 Nos			
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to as per ASME B16.34 APPENDIX II. & RMP61158 Rev 4 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.															
INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M VADIVEL - FICPL QA															
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements , material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.															
 M.Kahagalatha, Lab Chemist. Prepared by.										 C. Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT					

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:10440935 VSC/LAB/06 - R0			
INSPECTION CERTIFICATE AS PER EN 10204 3.1															
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/J7809/23 T.C.Date : 11-AUG-23					
P.Order : 10112521 Sino 7 Dt. 03/07/2023, 10111427 Sino 1 Dt. 09/05/2023, 10112521 Sino 2 Dt. 03/07/2023, 10112310 Sino 3 Dt. 21/06/2023										Heat No : J7809 Pour Date : 29-JUL-23					
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC					
CHEMICAL COMPOSITION % ()															
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080	
Actual	0.201	1.029	0.400	0.017	0.008	0.096	0.041	0.028	0.013	0.008	0.403	0.186	0.009	0.048	
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 5.00 Hrs and Air Cooled. Cycle No.D9236 Furnace ID.HT/02															
MECHANICAL PROPERTIES ()															
Requirements	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()						
	MPa	Ksi	MPa	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)						
	0.2 % Rp		-		-	-	-								
Minimum	290.00	42	485.00	70	22.00	35.00	---	-							
Maximum	---	---	655.00	95	---	---	237	-	1	2	3	Avg			
Actual	309.83	45	504.88	73	33.36	62.52	147,147,148		--	--	--	--			
ITEM DESCRIPTION															
SL.No.	Description					Drawing No.		Raw Part No/ Finishing Part No		CSL No		Quantity			
1	BODY CSTG 2" CL150					487845 REV 25		N487845WCC		5, 6		2 Nos			
2	BODY SN CAST 4" cl 150					2762516 REV 03		N2762516		3, 4		2 Nos			
3	BODY CSTG 2" cl 150					487845 REV 25		N487845WCC		7, 8, 9, 10		4 Nos			
4	BODY CSTG 2"cl 1500					487888 REV 39		N487888WCC		1, 2		2 Nos			
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to as per ASME B16.34 APPENDIX II. & RMP61158 Rev 4 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.															
 M VADIVEL - FICPL QA INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date: _____ Sample Qty: _____															
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements, material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.															
 M.Kahagalatha, Lab Chemist. Prepared by.										 C. Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT					

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:10440935 VSC/LAB/06 - R0			
INSPECTION CERTIFICATE AS PER EN 10204 3.1															
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/J7586/23 T.C.Date : 28-JUL-23					
P.Order : 10109417 Sino 8 Dt. 30/01/2023, 10111427 Sino 1 Dt. 09/05/2023, 10111186 Sino 1 Dt. 27/04/2023, 10110223 Sino 1 Dt. 09/03/2023										Heat No : J7586 Pour Date : 14-JUL-23					
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC					
CHEMICAL COMPOSITION % ()															
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080	
Actual	0.195	0.982	0.354	0.020	0.005	0.043	0.043	0.026	0.007	0.006	0.377	0.125	0.010	0.023	
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 4:00 Hrs and Air Cooled. Cycle No.D9195 Furnace ID.HT/02															
MECHANICAL PROPERTIES ()															
Requirments	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()						
	Ksi	Ksi	Ksi	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)						
	0.2 % Rp		-		-	-	-								
Minimum	42.06	42	70.34	70	22.00	35.00	---	-							
Maximum	---	---	95.00	95	---	---	237	-	1	2	3	Avg			
Actual	43.39	43	71.36	71	33.28	60.28	145,145,143		--	--	--	--			
ITEM DESCRIPTION															
SL.No.	Description				Drawing No.		Raw Part No/ Finishing Part No		CSL No	Quantity					
1	BODY SN CAST 2" cl 150				2765401 REV 04		N2765401		11	1 Nos					
2	BODY CSTG 2"cl 1500				487888 REV 39		N487888WCC		1, 2	2 Nos					
3	BODY SN CAST 2" cl.300				2765234 REV B		N2765234		9, 10	2 Nos					
4	BODY CSTG 2" cl 150				487845 REV 25		N487845WCC		3, 4, 5, 6, 7, 8	6 Nos					
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to ASME B16.34 APPENDIX II. & RMP61158 Rev 04 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.															
MTC - REVIEWED & ACCEPTED  10139322 FICPL QC															
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements , material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.															
 M.Kahagalatha, Lab Chemist. Prepared by.								 C.Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT							

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:10440935 VSC/LAB/06 - R0			
INSPECTION CERTIFICATE AS PER EN 10204 3.1															
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/J7571/23 T.C.Date : 28-JUL-23					
P.Order : 10110817 Sino 1 Dt. 07/04/2023, 10111427 Sino 1 Dt. 09/05/2023, 10111186 Sino 1 Dt. 27/04/2023										Heat No : J7571 Pour Date : 12-JUL-23					
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC					
CHEMICAL COMPOSITION % ()															
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080	
Actual	0.197	0.944	0.364	0.020	0.008	0.044	0.025	0.022	0.005	0.004	0.370	0.100	0.009	0.037	
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 6:00 Hrs and Air Cooled. Cycle No.D9180 Furnace ID.HT/02															
MECHANICAL PROPERTIES ()															
Requirements	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()						
	MPa	Ksi	MPa	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)						
	0.2 % Rp		-		-	-	-								
Minimum	290.00	42	485.00	70	22.00	35.00	---	-							
Maximum	---	---	655.00	95	---	---	237	-	1	2	3	Avg			
Actual	327.16	47	529.08	77	33.92	61.53	143,143,143		--	--	--	--			
ITEM DESCRIPTION															
SL.No.	Description				Drawing No.		Raw Part No/ Finishing Part No		CSL No	Quantity					
1	BODY SN CAST 2" cl 150				2765401 REV 04		N2765401		8	1 Nos					
2	BODY SN CAST 4" cl 150				2762516 REV 03		N2762516		4, 5, 6, 7	4 Nos					
3	BODY CSTG 2"cl 1500				487888 REV 39		N487888WCC		1, 2, 3	3 Nos					
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to ASME B16.34 APPENDIX II. & RMP61158 Rev 04 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.															
MTC - REVIEWED & ACCEPTED  10139322 FICPL QC															
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements , material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.															
 M.Kanagalatha, Lab Chemist. Prepared by.								 C. Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT							

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

Page 1 of 1

		Veeyes Steelcast Private Limited. 4/363-A,Kallipalayam,Chikarampalayam Village Post, Karamadai, Coimbatore, Pincode :641104										ISO CERTIFICATE No:10440935 VSC/LAB/06 - R0			
INSPECTION CERTIFICATE AS PER EN 10204 3.1															
Customer : FLOWSERVE INDIA CONTROLS PVT LTD-NVI CHENGALPATTU 603209										T.C. No : FS/J7494/23 T.C.Date : 22-JUL-23					
P.Order : 10109417 Sino 8 Dt. 30/01/2023, 10111427 Sino 1 Dt. 09/05/2023										Heat No : J7494 Pour Date : 07-JUL-23					
Specification : ASTM A216/A216M Gr.WCC -2021										CASTING PROCESSED BY INDUCTION MELTING FOUNDRY IDENTIFICATION : VSC					
CHEMICAL COMPOSITION % ()															
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RES	N	Al	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.015	0.080	
Actual	0.200	0.957	0.456	0.017	0.007	0.048	0.043	0.036	0.005	0.007	0.381	0.139	0.010	0.049	
HEAT TREATMENT : NORMALIZING - Heated to 920°C Soaked for 6:00 Hrs and Air Cooled.Cycle No.D9175 Furnace ID.HT/02															
MECHANICAL PROPERTIES ()															
Requirements	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness	Bend Test	Impact Test in ()						
	MPa	Ksi	MPa	Ksi	GL :50 mm	In Area	HBW		(10mm X 10mm X 55mm)						
	0.2 % Rp		-		-	-	-								
Minimum	290.00	42	485.00	70	22.00	35.00	---	-							
Maximum	---	---	655.00	95	---	---	237	-	1	2	3	Avg			
Actual	308.20	45	514.06	75	35.12	65.37	145,145,145		--	--	--	--			
ITEM DESCRIPTION															
SL.No.	Description					Drawing No.		Raw Part No/ Finishing Part No		CSL No		Quantity			
1	BODY CSTG 2"cl 1500					487888 REV 39		N487888WCC		1, 2, 3		3 Nos			
2	BODY CSTG 2" cl 150					487845 REV 25		N487845WCC		7, 4, 5, 6		4 Nos			
Remarks : Visual inspection as per MSS SP-55 No surface irregularities to Type I found, Surface irregularities to type II - XII do not exceed example 'a'. Radioactive level measured, delivered materials are free of radioactive contamination. MPI performed and acceptable to ASME B16.34 APPENDIX II. & RMP61158 Rev 4 The above part confirms to as per NACE MR0175 - 2021 / ISO 15156-2020. The above part confirms to as per RMC90012-REV 15. Carbon Steel Fully killed & fine Grained.															
MTC - REVIEWED & ACCEPTED  10139322 FICPL QC															
We Confirm that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements , material specification ASTM A216/A216M Gr.WCC -2021 and was found to meet the requirements.															
 M.Kahagalatha, Lab Chemist. Prepared by.										 C.Kannan, Manager QA. Makers Representative. SNT-TC-1A-LEVEL-II VT					

Heat COC

Desc 2"900#RF/NORDSTROM#2345 LUBRICATED DYNAMIC BALANCE,PLUG VALVE. CARBON STEEL BODY AND TRIM, METAL SEAT WITH DB-2 WRENCH OPERATOR

QA/F/44 12-2016

 SAMCO METALS AND ALLOYS PVT.LTD. ISO 9001 - 2015. TUV CERT.REG.NO:44 100 094292 AD-WO TUVNORD APPROVAL KANIYAMBADI,VELLORE Pin :632 102, Phone :0416-2230221, 2230222, 2917614, Fax :0416-2230459, email :samco@samcoalloys.in INDIA. CERTIFICATE OF MATERIAL AND TESTING FOR STEEL CASTING (As per EN 10204-3.1)															
Customer : FLOWERVE INDIA CONTROLS PVT.LTD B8, MMIDA INDUSTRIAL AREA , MARAIMALAI NAGAR -603209										T.C. No. : S4501/37 T.C.Date : 28/02/2023					
Reference. 10105198 Dt. 21/05/2022										Melt No. : S4501 Date of Pouring : 23/02/2023					
Specification : ASTM A216-2021 GR : WCC & RMC 90012 REV.14															
MELTING PROCESSED : BY BASIC INDUCTION															
Normalised : Raised to 940°C soaked for 5 hours and then air cooled. HEAT TREATMENT Details :															
SM :MANUFACTURER'S Symbol															
CHEMICAL COMPOSITION - %															
Element Requirement	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	N	TRE	CE	
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---	
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.080	0.015	1.000	0.430	
Actual	0.202	0.944	0.448	0.020	0.008	0.132	0.040	0.050	0.014	0.003	0.036	0.007	0.239	0.400	
MECHANICAL PROPERTIES															
Requirments	Yield Strength		Tensile Strength		%Elongation GL :50 mm	% Reduction of Area	Hardness HBW	Bend Test D = 3T at 90 °	Impact Test in (Joules) (10mm X 10mm X 55mm)						
	ksi	Kgf/Sq.mm	ksi	Kgf/Sq.mm					CVN 2mm 45 °						
		0.2% Proof Stress		-		-	-	-	-	-					
Minimum	40	28.05	70	49.47	22.00	35.00	---								
Maximum	---	---	95	66.81	---	---	237		1	2	3	Avg			
Actual	46	32.33	76	53.52	33.62	60.58	163,163	--	--	--	--	--			
DETAILS OF CASTING POURED															
SL.No.	Description					Drawing No.			Part No.		Quantity	Produced			
1	BODY CAST 2" CL 1500 F.NO.3045					487888 REV.AK			N487888WCC		17	17			
Remarks : MP No : MPW6817, MPW6818, MPW6819, MPW6820, MPW6821, MPW6822, MPW6823, MPW6824, MPW6825, MPW6826, MPW6827, MPW6828, MPW6829, MPW6830, MPW6831, MPW6832, MPW6833. We certify that the material was manufactured, sampled, tested and inspected in accordance with the material specification and the purchase order and was found to meet the requirements. The above castings are meet the specification FTDC/001 REV. 6. Castings have been MPI performed as per the FNDE - 002 REV. 07. Results found satisfactory. Visual inspection carried out as per MSS SP 55. No surface irregularities to type - I , Type II-XIIa - 2011. The above Castings are satisfactory Radiation results within limits of acceptable 0.20 μsv/hr.															
We Certify the particular entered herein are correct and the material satisfies the specification . ASTM A216-2021 GR : WCC & RMC 90012 REV.14															
 CHITRA.V (Manager QA)					MTC - REVIEWED & ACCEPTED  10139322 FICPL QC					for SAMCO METALS AND ALLOYS PVT.LTD.  JAGANNATHAN.S (Authorized Works Inspector)					